

Collaborative Teaching of Research Software Skills: Lessons from CodeRefinery for Higher Education and Research in Iceland

Hemanadhan Myneni and Ebba Þóra Hvannberg
Faculty of IMC, University of Iceland

Menntakvika 2026 · 30 September 2026

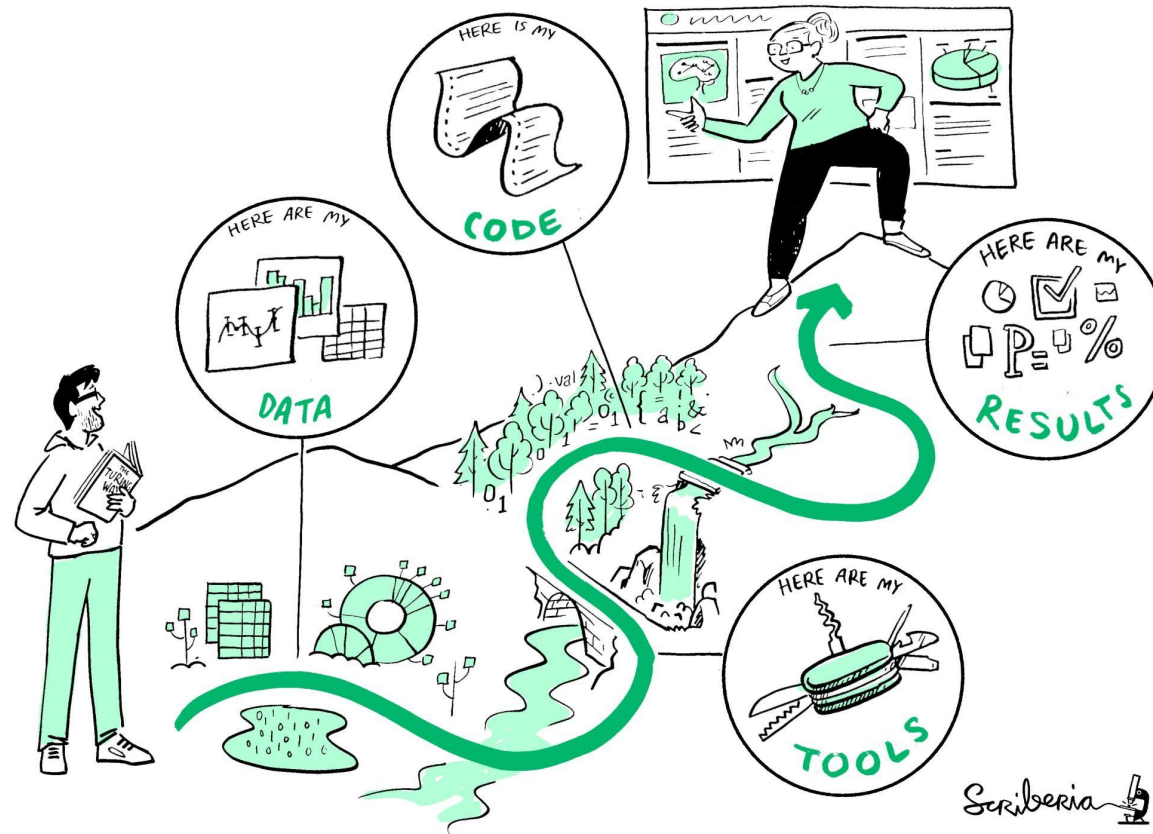


A Little About Me

- Research Specialist and Adjunct Associate Professor, Computer Science, University of Iceland
- Head, Quantum Simulation and Data Science Lab
- Physicist by training, computational researcher by practice
- Develop research software and computational workflows
- Teach across Physics and Computer Science — including quantum computing and HPC
- CodeRefinery member since 2025
- Board member, Nordic Research Software Engineers

Research Software in Everyday Research

Code, data, and workflows are part of the research method



Can
someone
retrace the
steps
from data to result?

Illustration: Scriberia for The Turing Way, CC BY 4.0

Research Software Skills: The Training Gap

Why these skills matter across research careers

Career stage

Master's student

PhD student

Researcher

Practical need

Trace changes to a thesis analysis and reproduce the results

Return to code months later and reliably reproduce previous results

Share methods and review changes with collaborators

The training gap

Many researchers acquire these skills informally, often outside formal degree programmes.

Carver et al., PeerJ Computer Science (2022); PLOS Computational Biology (2024)

What is Collaborative Teaching?

Educators plan and teach together, bringing complementary expertise to the same learning experience.

Shared planning · Complementary roles · Shared teaching

The key idea: no single instructor needs to do everything.

Collaborative Teaching in Practice

Different people contribute in different ways:

Lead instructor	introduces and explains concepts
Co-instructor	adds expertise and alternative explanations
Exercise helpers	support learners during hands-on work
Local hosts	connect streamed teaching with the local classroom

One course, shared across a teaching team.

Collaborative Teaching in Research Software Training

The collaborative teaching approach is established; its application to research software training is more recent.

- | | |
|-------------|---|
| 1998 | Software Carpentry begins training researchers in practical computing skills. |
| 2012 onward | Software Carpentry workshops expand through a community of instructors. |
| 2016 | CodeRefinery begins Nordic research software training. |
| Today | Instructors, helpers, and local facilitators collaborate across institutions. |

Collaborative teaching enables hands-on research software training to be shared across instructors, institutions, and locations.

Sources: Software Carpentry, *About Us*; CodeRefinery, *History of the Project*

What is CodeRefinery?

Training: Research software, reproducibility, open science & computational skills

Workshops: Typically offered in spring and autumn

Open & reusable: Teaching materials and recordings are available online;
most CodeRefinery-developed lessons are licensed under CC BY 4.0

CodeRefinery Workshop (22–24 September and 29 September–1 October 2026)

“Responsible use of generative AI in assisted coding” and more



CodeRefinery Course Offering in Iceland

Practical research software skills for working with code, data, and workflows

Course stage What participants practise

Days 1–2 Record changes, track their work, and return to an earlier version

Day 3 Share work and review a colleague's changes

Days 4–6 Reproducible research, social coding, documentation,
responsible use of generative AI in assisted coding, and automated testing.

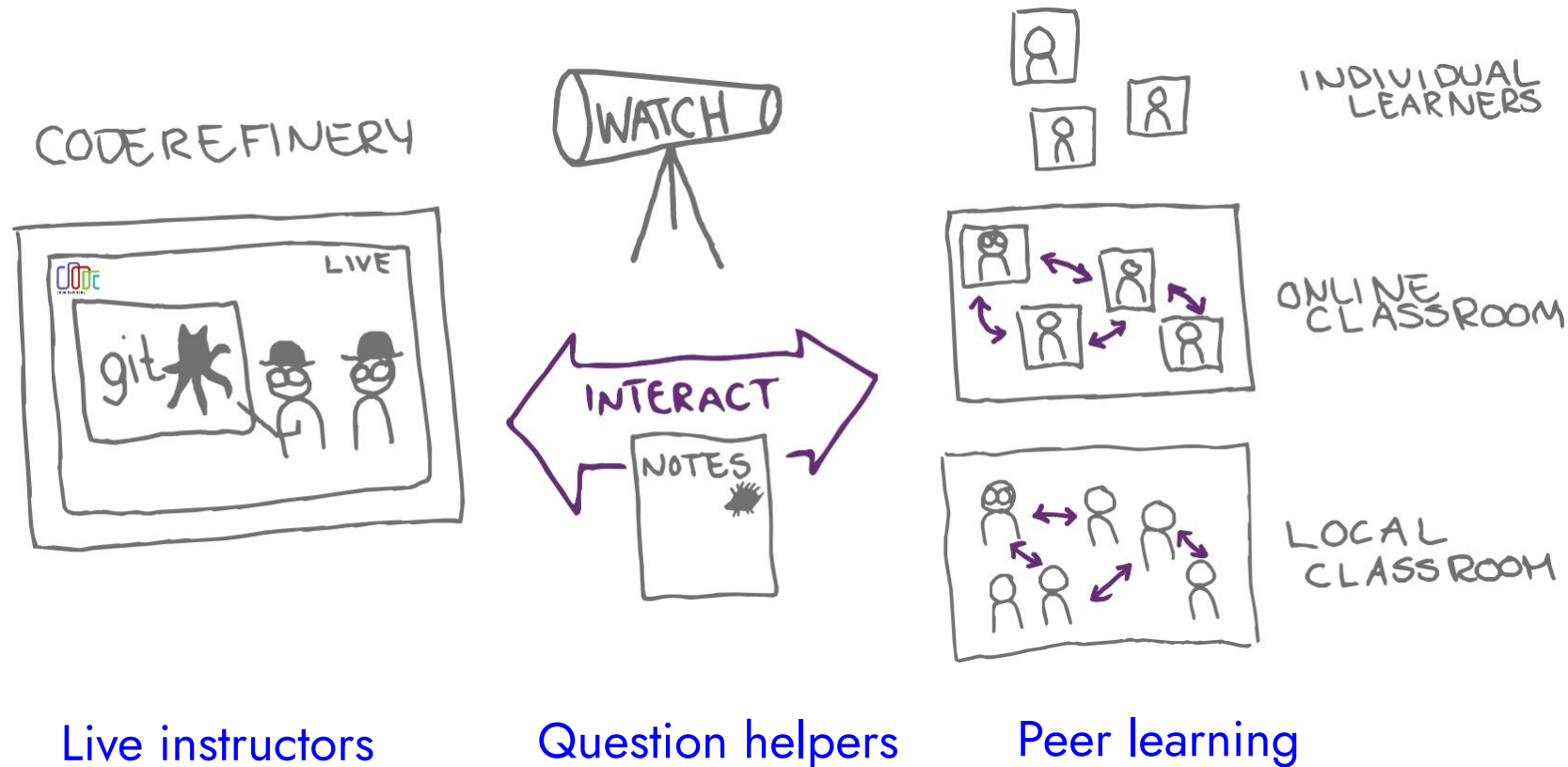
Who participates?

Students, researchers, research software developers, and research-support staff.

The Iceland classroom

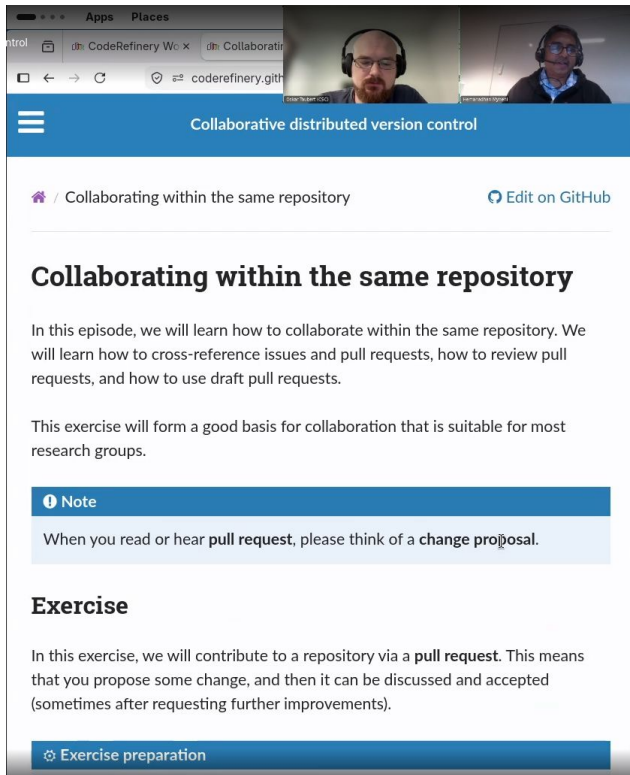
Live-streamed instruction with local, in-person support during hands-on exercises in Reykjavík.

CodeRefinery's Collaborative Teaching Model



The model enables teaching across institutions with support close to learners.

CodeRefinery's Collaborative Teaching in Practice



Collaborative distributed version control

/ Collaborating within the same repository [Edit on GitHub](#)

Collaborating within the same repository

In this episode, we will learn how to collaborate within the same repository. We will learn how to cross-reference issues and pull requests, how to review pull requests, and how to use draft pull requests.

This exercise will form a good basis for collaboration that is suitable for most research groups.

Note
When you read or hear **pull request**, please think of a **change proposal**.

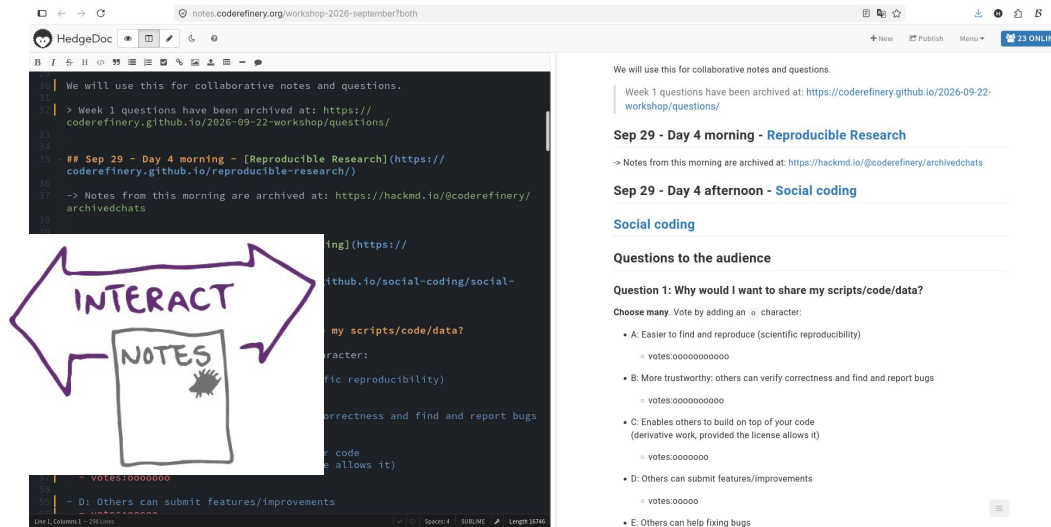
Exercise

In this exercise, we will contribute to a repository via a **pull request**. This means that you propose some change, and then it can be discussed and accepted (sometimes after requesting further improvements).

[Exercise preparation](#)



Watch online stream



HedgeDoc

notes.coderefinery.org/workshop-2026-september7both

We will use this for collaborative notes and questions.

Week 1 questions have been archived at: <https://coderefinery.github.io/2026-09-22-workshop/questions/>

Sep 29 - Day 4 morning - Reproducible Research

Notes from this morning are archived at: <https://hackmd.io/@coderefinery/archivedchats>

Sep 29 - Day 4 afternoon - Social coding

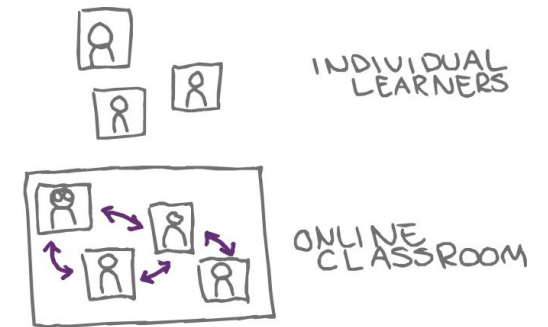
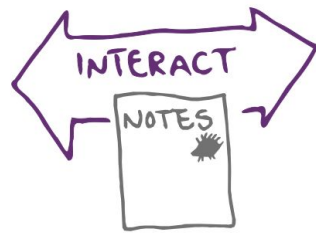
Social coding

Questions to the audience

Question 1: Why would I want to share my scripts/code/data?

Choose many. Vote by adding an @ character:

- A: Easier to find and reproduce (scientific reproducibility)
votes:oooooooooooo
- B: More trustworthy: others can verify correctness and find and report bugs
votes:oooooooooooo
- C: Enables others to build on top of your code (derivative work, provided the license allows it)
votes:oooooooooooo
- D: Others can submit features/improvements
votes:oooooo
- E: Others can help fixing bugs



Local classroom
University of Oslo Library

Live instructors

Question helpers

Peer learning

How CodeRefinery Applies the Idea

Two instructors One guides; another demonstrates or voices learner questions

Online helpers Answer questions in a shared document

Local facilitators Help a group work through live exercises

The model joins teaching across institutions with support close to learners.

Collaborative Teaching as a Conversation

Two ways to team teach live research software skills

Guide + demo-giver

Instructor 1: leads the lesson

Instructor 2: plays the learner role

Presenter + interviewer

Instructor 1: leads the lesson

Instructor 2: asks questions as a learner might ask

Agree who leads each segment and how to handle questions.

Source: CodeRefinery Train the Trainer, "Co-teaching"

CodeRefinery Workshops: Registration Statistics

Workshop	Total Registrations	Registrations from Iceland
Sept - Oct 2025	505	17
March 2026	740	27

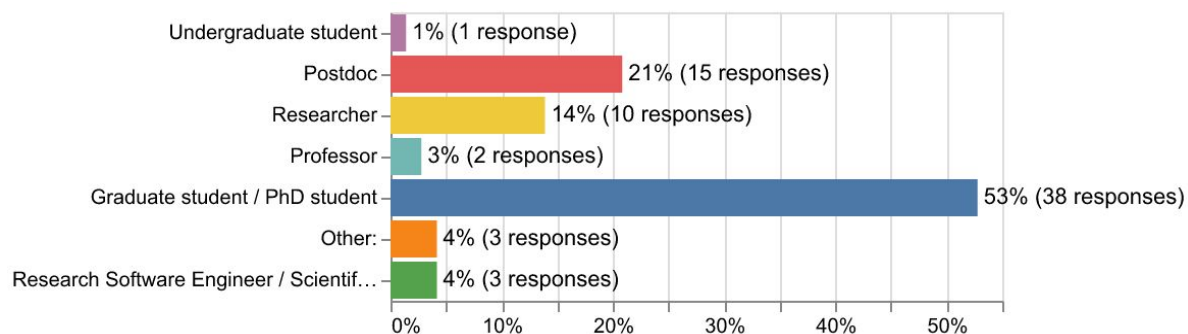
Total registrations increased by ~47%

Registrations from Iceland increased by ~59%

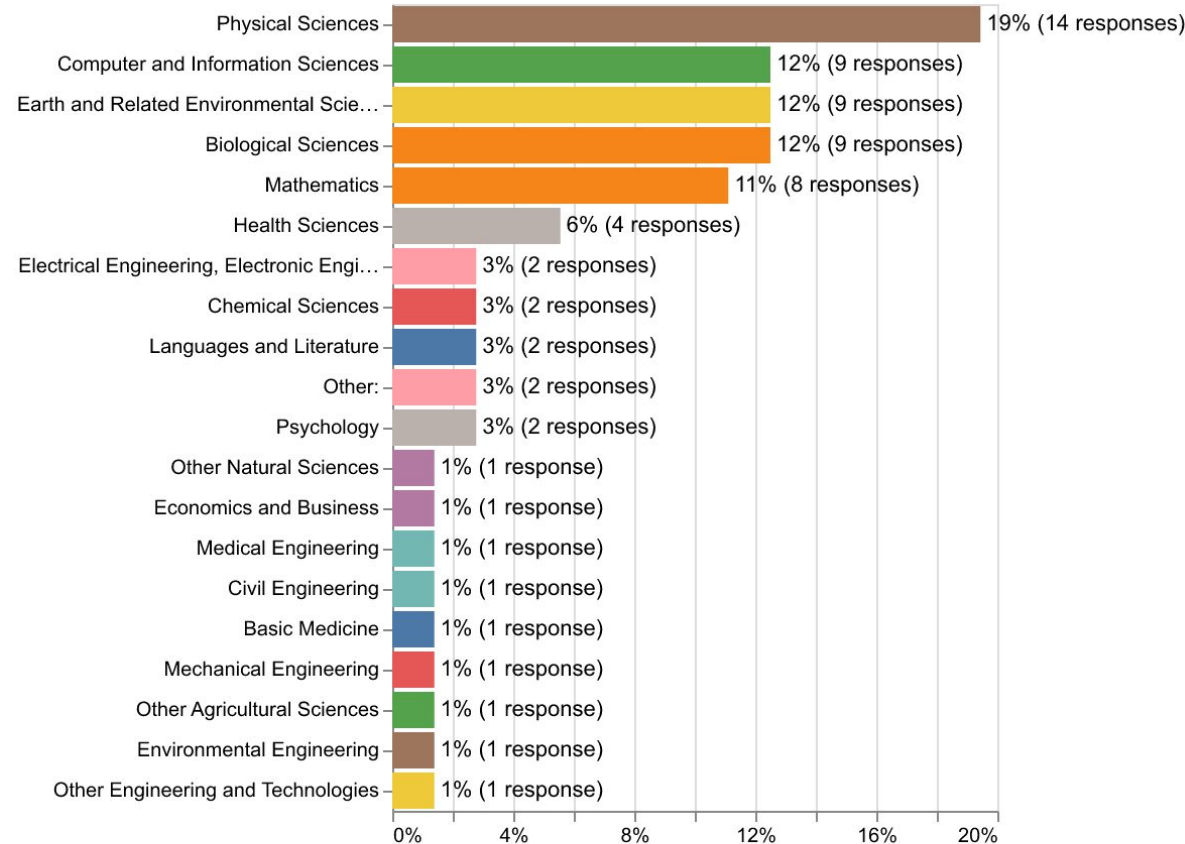
Registrations from Iceland include participants from HÍ, HR, and Hafró

CodeRefinery Workshops: Participants Career Stage and Academic Discipline

2026 post-workshop survey · 72 responses



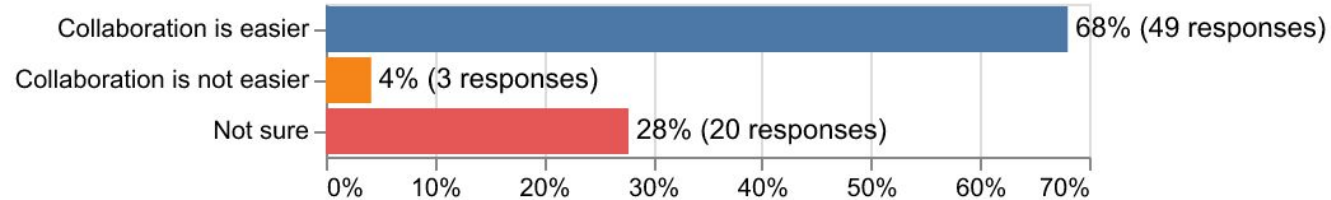
Career stage



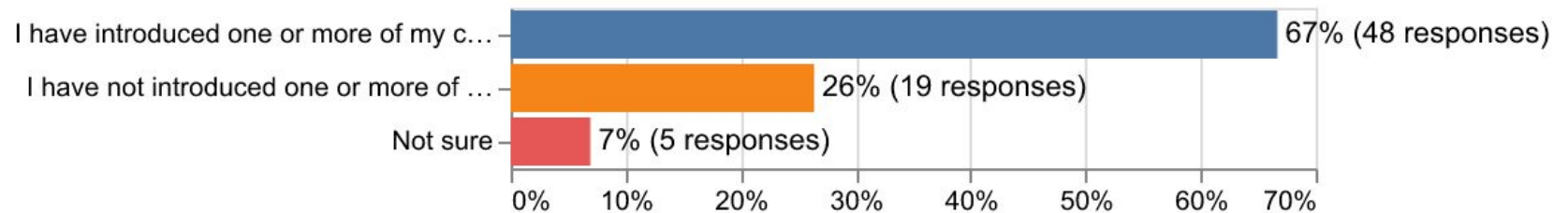
Academic discipline

2026 CodeRefinery Post-Workshop Survey

Participants from the 2024-2025 workshops · 72 responses



After attending the workshop, has it become easier or not for you to collaborate on software development with your colleagues and collaborators?



Have you introduced one or more of your colleagues to new tools or practices as a result of the workshop?

Future Work

- Explore integrating CodeRefinery into graduate researcher training at the University of Iceland.
- An example within Aurora: Vrije Universiteit Amsterdam offers the CodeRefinery workshop as a PhD course (1 ECTS) across its graduate schools.
- Adapt and reuse openly licensed CodeRefinery materials for local graduate training.

CodeRefinery Resources

- [Website](https://coderefinery.org/): <https://coderefinery.org/>
- [September–October 2026 Workshop](https://coderefinery.github.io/2026-09-22-workshop/) — Workshop materials & schedule
<https://coderefinery.github.io/2026-09-22-workshop/>
- [Live Workshop Stream](https://www.twitch.tv/moderator/coderefinery/) — Twitch
<https://www.twitch.tv/moderator/coderefinery/>
- [Recordings](https://www.youtube.com/@coderefinery) — YouTube
<https://www.youtube.com/@coderefinery>
- [Community & discussion](https://coderefinery.zulipchat.com/) — Zulip
<https://coderefinery.zulipchat.com/>
- [Live Q&A / collaborative notes](#) — HackMD



Takk fyrir 🇮🇸 Thanks



Contact

Hemanadhan MYNENI

myneni@hi.is